



FusRock™ FDM 3D Printing Material Technical Data Sheet

Data / Revised: 11.2025

Version No: 1.0

FusFlex™ TPU-GF

15%短切玻璃纤维增强 TPU

A high toughness and 78D shore hardness flexible 3D printing material

产品介绍

Product Description

FusFlex™ TPU-GF 是一款 15%短切玻璃纤维填充的半硬质 TPU 3D 打印材料，在保证一定柔韧性的同时兼顾了较好的刚性。FusFlex™ TPU-GF 具有极佳的抗冲击性能和耐磨性，也具有较高的耐热性。FusFlex™ TPU-GF 易于打印，可兼容大部分挤出机，可应用于部分替代 ABS 和高抗冲要求零部件生产。

FusFlex™ TPU-GF is a semi-rigid TPU 3D printing material filled with 15% chopped glass fibers, balancing good rigidity while maintaining a certain degree of flexibility. FusFlex™ TPU-GF offers excellent impact resistance and wear resistance, along with high heat resistance. It is easy to print, compatible with most extruders, and can be used in applications that partially replace ABS and in the production of components with high-impact resistance requirements.

产品详情

Available

颜色 Color: 白色 White/ 黑色 Black

线径 Diameter: 1.75mm

净重 Net Weight: 1KG

物性表

Material Properties

测试项目 Property	测试方法 Testing method	典型值 Typical value
密度 Density	ISO 1183	1.32 g/cm³
硬度 Hardness	ISO 7619	67 Shore D



熔融指数 Melt index		230°C, 2.16kg	22 g/10min
维卡软温度 Vicat softening temperature		ISO 306	157.3 °C
拉伸断裂强度 (X-Y) Tensile breaking strength (X-Y)		ISO 527	30.92±0.33 MPa
杨氏模量 (X-Y) Young's Modulus			429.80±15.82 MPa
断裂伸长率(X-Y) elongation at break (X-Y)			74.80±7.80 %
弯曲强度 (X-Y) Bending strength		ISO 178	14.47±0.36 MPa
弯曲模量 (X-Y) Bending Modulus			395.89±44.44 MPa
缺口冲击强度 (X-Y) Charpy impact strength		ISO179	未冲断 Non-break
Z 方向撕裂强度 (Z) Tear Strength (Z)	Cooling Fan 0%	ISO 34-1	63.21±5.89 KN/m
	Cooling Fan 50%		56.28±3.07 KN/m

试样打印参数：喷嘴大小 0.4mm，喷嘴温度 260°C，底板加热 30°C，打印速度 120mm/s，填充率 100%，填充角度±45°

Specimens printed under the following conditions: Nozzle size 0.4mm, Nozzle temp 260°C, Bed temp 30°C, Print speed 120mm/s, Infill 100%, Infill angle ±45°



建议打印参数

Recommended printing conditions

喷头温度 Nozzle temperature	240-270 °C
建议喷嘴大小 Recommended nozzle diameter	≥0.4mm
建议底板材质 Recommended build surface	玻璃, PEI 膜或 PC 膜 Glass, PEI Film or PC Film
底板温度 Build plate temperature	40-60°C
Raft 间距 Raft separation distance	0.18-0.22mm
冷却风扇 Cooling fan speed	20-50%
打印速度 Print speed	30-120 mm/s
回抽距离 Retraction distance	0.8-1.0 mm
回抽速度 Retraction speed	1800-3600 mm/min

其他建议:

Additional Suggestions:

1. 如果想实现高速打印, 推荐使用近程挤出机, 例如: **BMG** 挤出机、**Titan** 挤出机、**Hemera** 挤出机, 并适当提高喷嘴温度。

If you want to achieve high-speed printing, it is recommended to use direct extruders, such as BMG extruder, Titan extruder and Hemera extruder, and appropriately increase the nozzle temperature.

2. TPU 材料暴露在空气中容易吸收水分, 吸湿后打印会出现拉丝, 挤出有气泡, 打印表面粗糙等现象, 降低打印质量。建议您打开 **FusFlex™ TPU-GF** 真空铝箔袋包装后立即将线材放入干燥盒内 (湿度控制在 15%以下) 进行打印。不用的线材请放回原包装铝箔袋内密封保存。

TPU material is very easy to absorb moisture when exposed to air, and printing after absorbing moisture will result ozzing, extruding with bubbles and rough surface appearance, thus reducing print quality. It is



recommended that put the filament into a dry box (humidity below 15%) immediately after opening the FusFlex™ TPU-GF vacuum foil bag for printing. Please put the unused filament back into the original aluminum foil bag for sealed storage.

3. 材料受潮后会出现打印拉丝增多，挤出有气泡，打印表面质量粗糙等现象。请将线材放入 70-80°C 烘箱内干燥 4-6h，即可恢复 FusFlex™ TPU-GF 的打印质量。

After the material is damp, there will be more printing oozing, bubbles extruded and rough printing surface. Please dry the filament in an oven at 70-80°C for 4-6h to restore the printing quality of FusFlex™ TPU-GF.

免责声明

Disclaimer

Fusrock® 3D 打印耗材适用于通用打印用途，已在标准条件下进行测试。然而，打印成品的性能与安全性受多种因素影响，包括打印参数、模型设计、使用环境及实际用途。使用 Fusrock® 材料即表示用户已知悉并同意，自行评估打印件是否适用于其具体应用，并承担由此产生的全部风险。Fusrock® 对使用本公司耗材打印的产品在实际应用中可能导致的任何损害、伤害或损失不承担任何责任，包括但不限于结构失效、功能异常或使用环境中的安全隐患。在将打印件应用于关键、功能性或商业性场景前，请务必进行充分测试。除 Fusrock® 已标明材料所获得的各项认证资质外，Fusrock® 产品未取得医疗、航天或生命支持系统认证资质。

Fusrock® 3D printing filaments are suitable for general printing applications and have been tested under standard conditions. However, the performance and safety of printed products are influenced by multiple factors, including printing parameters, model design, operating environment, and intended use. By using Fusrock® materials, users acknowledge and agree to independently evaluate the suitability of printed items for their specific applications and assume all associated risks. Fusrock® shall not be held liable for any damages, injuries, or losses resulting from the practical use of products printed with its materials, including but not limited to structural failures, malfunctions, or safety hazards in operational environments. Thorough testing must be conducted before applying printed components to critical, functional, or commercial scenarios. Fusrock® products are not certified for medical, aerospace, or life-support systems, except for certifications explicitly stated by Fusrock® for specific materials.